Universal gravitational energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the zero-at-infinity convention for gravitational potential energy.

SUCCESS CRITERIA - I can...

1. Use center-to-center distance.
2. Explain why negative potential energy does not mean negative kinetic energy.
3. Compute changes from final minus initial energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “potential energy” and “radius” to explain your answer. Sentence starters are optional.

potential energy: energy associated with the configuration of interacting objects

radius: distance from the center to the edge of a circle or sphere

initial: at the beginning of the interval; subscript i

01 Warm-up: bring back an earlier idea

Net force rises linearly from zero to **10 N** over **0.3 m**. **Find work** and final speed for a **0.3 kg** cart starting at rest.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

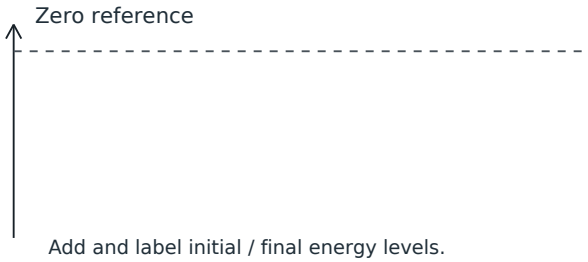
02 Worked example: explain the prediction

Reference: $U = -GMm/r$; $\Delta U = U_f - U_i$.
Outside a spherical source; **choose U = 0** at infinity. Moving outward makes U less negative.

Worked reasoning: For a supplied source with GMm=**120 J·m**, potential energy is U=**-120**/r. At r=**3 m**, U=**-40 J**; at **6 m**, U=**-20 J**. Moving outward raises U by **20 J**.

Your turn: Draw an energy number line and **mark the direction toward the** zero at infinite separation.

Sentence starter: In my work, I used … to show … . I labeled it … .



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

An object moves outward from r to **2** r. Does gravitational potential energy become more or less negative? **Draw a number-line comparison**.

Sentence starter: In my work, I used … to show … . I labeled it … .





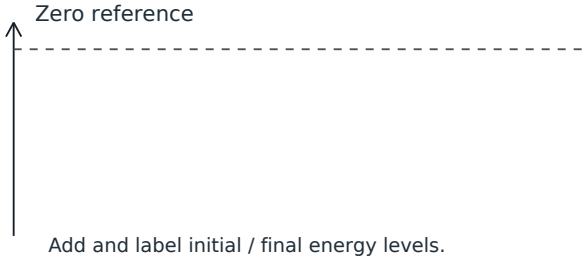
Example: potential energy (J)

Marzano check - circle: ① ② ③

04 Apply the physics

A **1000 kg** satellite moves from r=**7.0×10^6** to **1.4×10^7 m**.
1. Use GM=**4.0×10^{14} m^3/s^2**.
2. **Find U** at each point and ΔU.

Sentence starter: I found … [unit]. This value means … in this situation.



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says “negative potential energy means the object has negative kinetic energy.” Correct the statement.

Sentence starter: I notice … . This supports … because … .

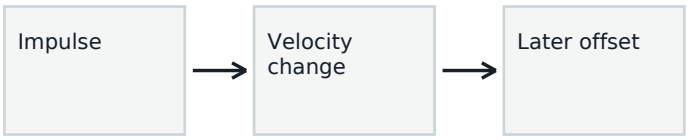
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A hypothetical target needs a **4 km** sideways shift over **6×10⁶ s**. Its mass is **8×10⁸ kg**. Estimate Δv and impulse; **describe the calculation you used** and what it treats as true.
Label the quantities in the diagram and say what it treats as true.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

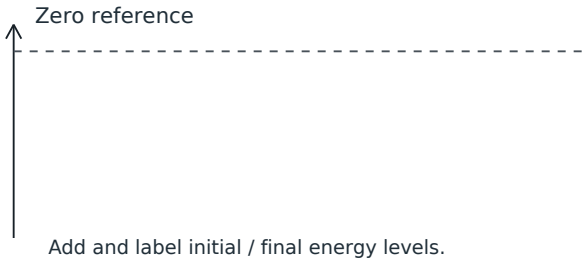


Marzano check - circle: 1 2 3

07 Exit check

With GMm=**60 J·m**, move from r=**5** to r=**15 m**. **Find work** by gravity.

Sentence starter: I found … [unit]. This value means … in this situation.



Add and label initial / final energy levels.

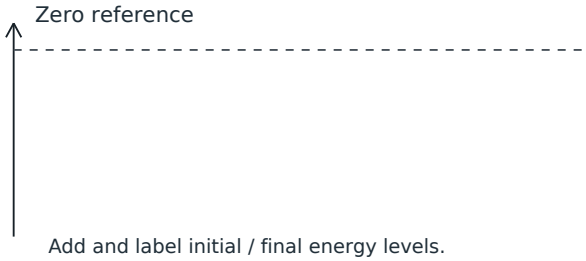
Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A mass moves from r=R to **3** R. Express ΔU and work by gravity in terms of GMm/R. Compare with reaching infinity.

Sentence starter: Both … . They differ in … because … .



Add and label initial / final energy levels.

Label energies relative to U = 0 at infinite separation.

Marzano check - circle: 1 2 3